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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
First SEMESTER

Discipline Specific Course (DSC) - MG1DSCFSC100 – GENERAL FORENSIC SCIENCE
(2024 ADMISSION ONWARDS)

PART A – Short Answer Type Questions (Any 10 × 2 = 20 Marks)

1. Branch of forensic science dealing with chemical analysis of crime exhibits. (1 mark)
   Applications: detection of poisons, explosives, drugs, adulterants, arson residues, inks, dyes, toxins. (1 mark)

2. Professionals who analyse physical/chemical/biological evidence for law/justice. (1 mark)
   Role: examination, interpretation, report writing, court testimony. (1 mark)

3. Out of syllabus – Any relevant attempt may give full marks

4. On-site analysis of digital devices, Rapid evidence collection, Reduces contamination/tampering; improves efficiency. (Any two points -2 Marks)

5. The law of circumstantial facts in forensic science refers to the principle that "facts don't lie, but people can". It emphasizes that physical and scientific evidence (circumstantial facts) is more reliable for investigation than human statements, as it can be analyzed to establish links between a suspect and a crime, even if indirectly. (2 marks)

6. Forensic Biology primarily associated with analysis of blood, semen, saliva, hair, tissues. Helps by DNA profiling to link victim, suspect, crime scene.(1+1 mark)

7. Forensic Psychology - Behavioural profiling, assessment of mental state. Helps in suspect identification, interrogation, offender profiling. (1+1 mark)

8. Director → Joint/Deputy Director → Assistant Directors (Biology, Chemistry, Ballistics, Toxicology, Documents, Cyber) → Scientific officers→ Technicians. Units work under administrative & technical hierarchy. (2 Marks)

9. Juvenile shelter home- Institution for care, protection, and rehabilitation of children in conflict with law or in need of care. (2 Marks)
10. Freedom of movement, no high walls. Encourages rehabilitation and social reintegration. Reduces overcrowding. (Any relevant two points 2 Marks)

11. International standard for competence of testing & calibration laboratories.- Specifies quality management and technical requirements. (2 Marks)
12. Legal process of charging and trying an accused in court, Conducted by Public Prosecutor on behalf of the state.

PART B – Short Essay (5 × 4 = 20 marks)

13. The Law states that *everything undergoes change over time*—biological, chemical, and physical. In forensic science, this means evidence deteriorates (blood dries, fingerprints fade, bodies decompose), so timely collection is crucial.It guides investigators to document and preserve evidence before environmental or human factors alter it. (2 marks explanation + 2 marks establishing relation)

14. Evidence includes all materials—physical, documentary, digital, biological—presented before a court to establish facts. (2 marks)
    It must be relevant, admissible, reliable, and legally obtained. Chain of custody and expert testimony support its credibility. Evidence guides judicial decision-making. (2 marks)

15. Explanation of role of media in criminal investigation
    a. May influence public opinion, witness statements.
    b. Can cause pressure on investigators → "trial by media".
    c. Could help by generating leads, but risks misinformation.
    d. Affects fairness of trial. Any valid four points – 4 marks

16. Role of judicial officers include
    1. Issue search/arrest warrants.
    2. Authorise remand, oversee legality of investigation.
    3. Evaluate investigation reports.
    4. Ensure protection of rights of accused and victim. Any valid four points – 4 marks

17. A forensic scientist examines physical evidence, performs laboratory tests, interprets results, and prepares expert reports. (Forensic expert – 1 mark)
    They assist police in crime reconstruction, ensure scientific accuracy, maintain chain of custody, and give impartial testimony in court.
    They uphold quality and ethical standards in investigations. (3 valid points – 3 mark)
18. Forensic ballistics studies firearms, ammunition, and projectile behavior.
    Bullet striations, firing pin marks, breech face impressions, and cartridge case marks are compared using comparison microscopes. (Forensic ballistics – 2 mark)
    These patterns can uniquely match a bullet to a specific weapon.
    Residue patterns help reconstruct firing distance and shooter position. (2 valid points – 2 mark)

19. Law enforcement plays a central role through patrol, investigation, intelligence, and public safety actions. (Law enforcement- 2 marks)
    Challenges include limited manpower, outdated technology, cybercrime growth, case backlog, corruption, and lack of coordination.
    Reforms in training, forensic adoption, digitalization, and community policing can strengthen effectiveness. (2 valid points – 2 mark)

PART C – Essay Type Questions (Any 1 × 10 = 10 Marks)

20. **Definition:**
    Forensic science is the application of scientific principles and laboratory techniques to analyze crime-related evidence for use in courts of law. (2 marks)

Significance:
Ensures objective and scientific evaluation of evidence.
Helps identify victims and suspects.
Reconstructs crime scenes and establishes timelines.
Enhances reliability of investigations and court judgments. (3 marks)
High-profile examples:
Nirbhaya Case (2012): DNA profiling and digital evidence identified all accused.
Aarushi-Hemraj case: Forensic odontology, blood spatter analysis, and scene reconstruction were crucial. (Detailed explanation 5 marks)

21. **Crime scene:**
    A location where an offense has occurred or where evidence related to the crime is present. (2 marks)

Crime Scene Management Steps:
Securing the scene: First responders protect life and isolate the area.
Documentation: Photography, videography, sketching, and note-taking.
Systematic search: Grid/spiral/zone methods to locate evidence.
Evidence collection: Proper handling, packaging, labeling, and chain of custody.
Preservation: Prevent contamination or loss using PPE and sealed containers.
Transport and storage: Evidence sent to laboratory safely.
Crime scene reconstruction: Investigators interpret clues to recreate the sequence of events. (6 marks)
Importance:
Proper management ensures accuracy, prevents tampering, and strengthens the evidentiary value in court. (2 marks)